

# Prime Throughput Performance Evaluation

## Report 2

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### Source Code and GitHub Repositories

The files used for this evaluation can be found at the following repository:

<https://github.com/sam444555/cs2910>

The original Prime code used is available at the following Johns Hopkins DSN repository:

<https://github.com/jhu-dsn/prime>

### Testing Environment Issues, Resolution, Delays, and Next Week's Objectives

Following the testing documented in the first [Prime Throughput Performance Evaluation](#), a Windows OS update caused issues with the previously functioning Docker environment. A throughput test which originally peaked at 1042 updates/sec was now peaking at 42 updates/sec, or a decrease in throughput performance of about 96%.

After a substantial amount of troubleshooting, I was able to determine the cause as Hyper-V's TimeSync incorrectly applying time corrections to WSL2's system clock.

- Microsoft's Hyper-V is the virtualization technology which enables Linux-based Docker containers to run on Windows using WSL2 (Windows Subsystem for Linux 2)
- WSL2 provides a Linux environment (Ubuntu 22.04 LTS) within a virtual machine hosted on Windows where Docker containers share a common Linux kernel

This issue caused the environment's wall clock to incorrectly be adjusted by 5.4-5.8 seconds forward. Ubuntu's NTP (Network Time Protocol) recognized this time discrepancy and attempted to correct the system clock. This resulted in a tug of war between NTP and Hyper-V's time sync mechanism, which ultimately resulted in the degradation of Prime/Spines performance. I have found a solution to this issue, which can be found [here](#) in the README.

I spent a decent chunk of the week troubleshooting this issue which limited the amount of further work I planned on having done. This includes:

- Running the configuration on Aster
- Driver modifications/optimizations

- Throughput testing with encryption enabled in Spines
- More extensive Multi-driver testing

After having resolved the issue, I went back and reran and extended bandwidth-limited tests from the first Throughput Performance Evaluation (no RTT emulation) to verify the original results had not been adversely affected by this timing bug. These tests are the **Single-Driver Configuration** tests which include varying network speeds of 100/250/500/750 Mbps across 10/15/20/25/30/50/100 emulated clients. I have included the latency numbers as well, in milliseconds, as we had discussed in our previous meeting. The testing confirmed that the timing bug was not present in the first throughput performance tests. I am still not 100% sure what triggered the bug, but I was able to restore the environment to its original state.

I then performed limited **Multi-Driver Configuration** testing which had the driver process running across all six containers with the update workload equally distributed amongst them. Due to time constraints this testing was not as extensive as I had hoped it would be, however what data I did up getting proved interesting and warrants further testing.

My main focus for this upcoming week is to identify and eliminate the cause of the sending failures and stalls still present when running the driver program, with the goal of establishing a more robust and reliable testing environment. Interestingly, after resolving the timing issue, the frequency of sending failures and stalls decreased relative to the first throughput tests. This raises the possibility that this behavior may be environment influenced (OS choice, virtualization, etc) and suggests the need for testing across different environments. It was not completely eliminated, however, tests with higher *Number\_Of\_Emulated\_Clients* (recall this is the maximum number of outstanding updates) would still stall indefinitely at times. This issue also persisted in the Multi-Driver configuration, requiring multiple test attempts to obtain the data set.

Beyond this, I plan to make progress on the aforementioned work I was unable to complete this week. Additionally, I would like to streamline the Docker environment/setup and incorporate functionality that allows for easy switching between the local Docker setup and the Aster cluster.

## Single-Driver Configuration Testing

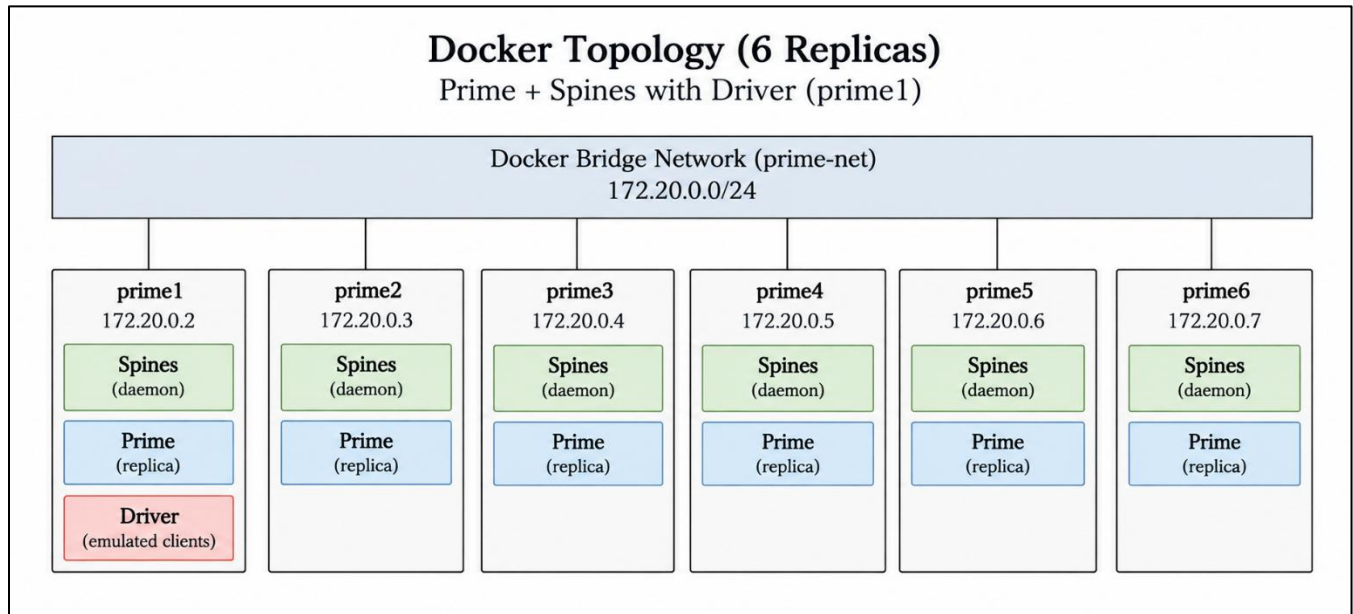


Figure 1. The first series of throughput testing will have a single driver program executing all the updates in a single container. (total updates =25k)

The single-driver configuration has a single driver program executing on the prime1 container.  
The workload of 25k updates is handled by this container only.

### 100 Mbps Link Speed (total updates = 25k)

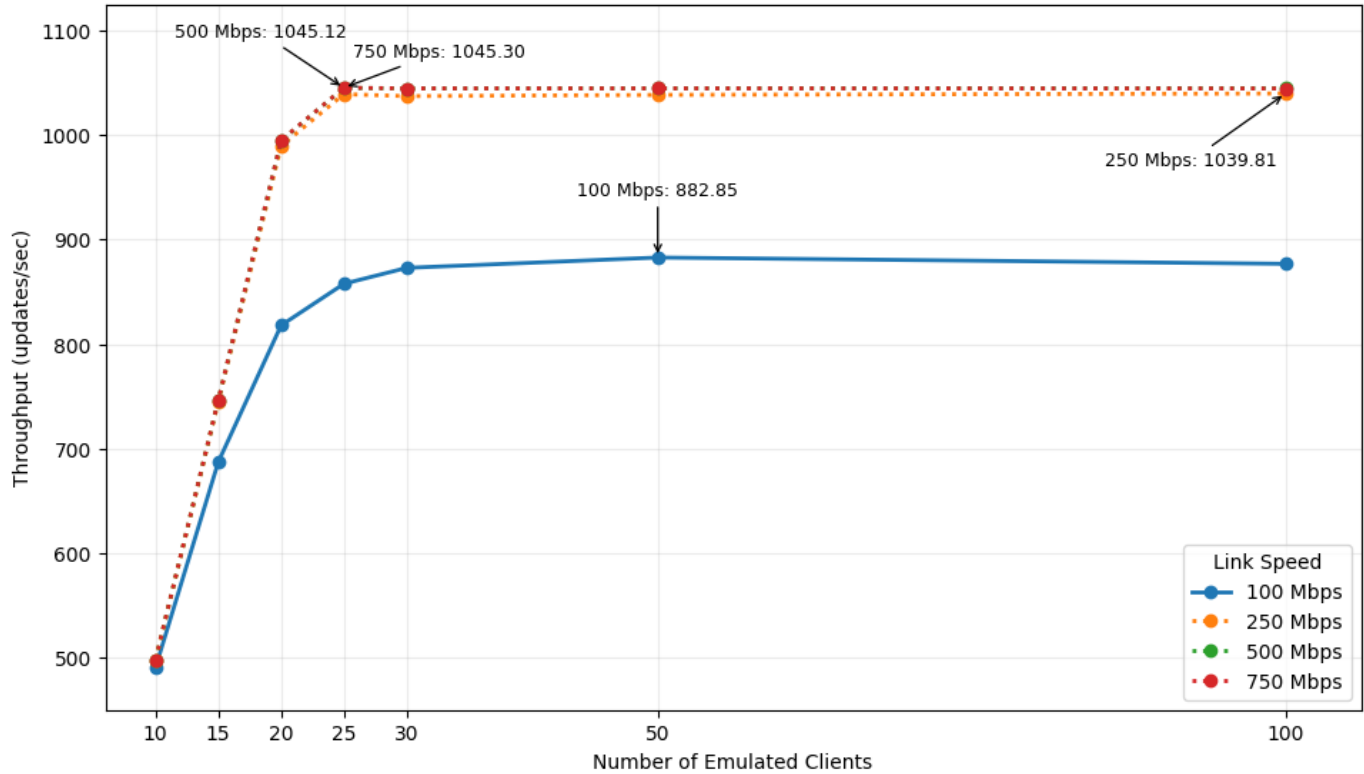
| Emulated Clients | Throughput (updates/sec) | Time Elapsed (secs) | Failed Sends | Min Latency (ms) | Max Latency (ms) | Avg Latency (ms) |
|------------------|--------------------------|---------------------|--------------|------------------|------------------|------------------|
| 10               | 490.59                   | 50.96               | 0            | 18.03            | 43.331           | 20.38            |
| 15               | 687.29                   | 36.38               | 0.67         | 16.382           | 41.114           | 21.818           |
| 20               | 817.91                   | 30.57               | 2.6          | 17.212           | 43.733           | 24.444           |
| 25               | 857.7                    | 29.18               | 575.6        | 18.457           | 60.224           | 29.155           |
| 30               | 872.95                   | 28.64               | 5135.8       | 18.745           | 67.968           | 34.21            |
| 50               | 882.85                   | 28.35               | 24968.6      | 18.724           | 78.412           | 37.515           |
| 100              | 876.84                   | 28.52               | 24965        | 19.401           | 75.938           | 37.736           |

### 250 Mbps Link Speed (total updates = 25k)

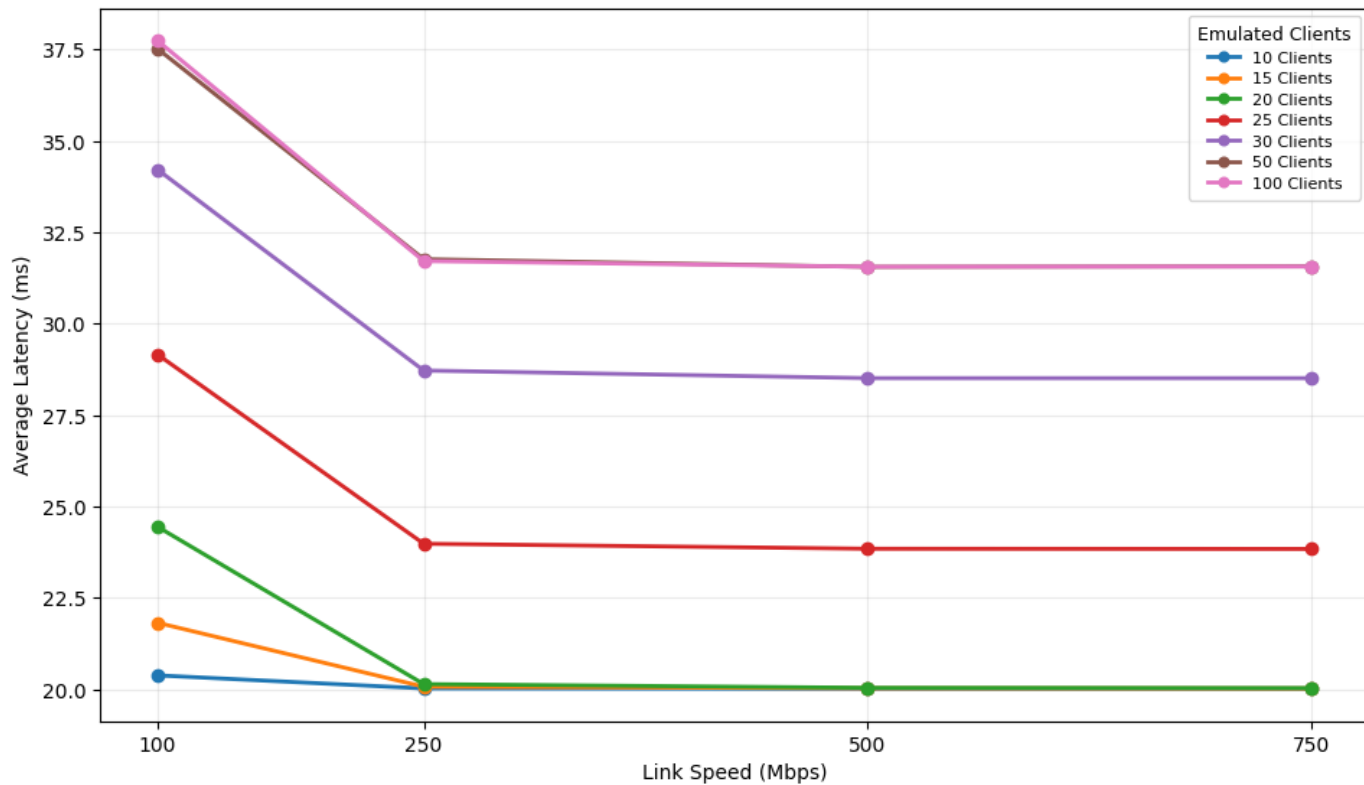
| Emulated Clients | Throughput (updates/sec) | Time Elapsed (secs) | Failed Sends | Min Latency (ms) | Max Latency (ms) | Avg Latency (ms) |
|------------------|--------------------------|---------------------|--------------|------------------|------------------|------------------|
| 10               | 497.78                   | 50.22               | 0            | 16.646           | 40.224           | 20.023           |
| 15               | 745.01                   | 33.56               | 1.4          | 18.103           | 40.127           | 20.069           |
| 20               | 989.6                    | 25.26               | 4            | 16.407           | 42.333           | 20.141           |
| 25               | 1039.17                  | 24.06               | 1473.6       | 17.405           | 52.992           | 23.979           |
| 30               | 1037.24                  | 24.1                | 7359.8       | 18.436           | 61.218           | 28.715           |

|                                           |                                 |                            |                     |                         |                         |                         |
|-------------------------------------------|---------------------------------|----------------------------|---------------------|-------------------------|-------------------------|-------------------------|
| 50                                        | 1038.46                         | 24.08                      | 24968.8             | 18.816                  | 66.541                  | 31.76                   |
| 100                                       | 1039.81                         | 24.04                      | 24966               | 17.1                    | 59.699                  | 31.708                  |
| 500 Mbps Link Speed (total updates = 25k) |                                 |                            |                     |                         |                         |                         |
| <b>Emulated Clients</b>                   | <b>Throughput (updates/sec)</b> | <b>Time Elapsed (secs)</b> | <b>Failed Sends</b> | <b>Min Latency (ms)</b> | <b>Max Latency (ms)</b> | <b>Avg Latency (ms)</b> |
| 10                                        | 497.92                          | 50.21                      | 0                   | 15.898                  | 38.554                  | 20.015                  |
| 15                                        | 746.46                          | 33.49                      | 0.2                 | 17.975                  | 39.737                  | 20.026                  |
| 20                                        | 994.72                          | 25.13                      | 3                   | 17.653                  | 39.279                  | 20.039                  |
| 25                                        | 1045.12                         | 23.92                      | 1480.8              | 18.051                  | 47.634                  | 23.842                  |
| 30                                        | 1044.66                         | 23.93                      | 7412.2              | 18.39                   | 56.408                  | 28.506                  |
| 50                                        | 1044.77                         | 23.93                      | 24962.6             | 18.351                  | 50.834                  | 31.549                  |
| 100                                       | 1044.88                         | 23.93                      | 24964.8             | 18.543                  | 53.303                  | 31.552                  |
| 750 Mbps Link Speed (total updates = 25k) |                                 |                            |                     |                         |                         |                         |
| <b>Emulated Clients</b>                   | <b>Throughput (updates/sec)</b> | <b>Time Elapsed (secs)</b> | <b>Failed Sends</b> | <b>Min Latency (ms)</b> | <b>Max Latency (ms)</b> | <b>Avg Latency (ms)</b> |
| 10                                        | 497.99                          | 50.2                       | 0                   | 16.437                  | 37.503                  | 20.016                  |
| 15                                        | 746.68                          | 33.48                      | 0.4                 | 17.761                  | 39.535                  | 20.021                  |
| 20                                        | 995.04                          | 25.13                      | 6.2                 | 16.871                  | 39.424                  | 20.031                  |
| 25                                        | 1045.3                          | 23.91                      | 1477                | 18.27                   | 46.14                   | 23.837                  |
| 30                                        | 1044.66                         | 23.93                      | 7419.6              | 18.612                  | 53.786                  | 28.506                  |
| 50                                        | 1044.69                         | 23.93                      | 24974               | 18.452                  | 55.25                   | 31.562                  |
| 100                                       | 1044.59                         | 23.93                      | 24966.8             | 16.32                   | 59.57                   | 31.559                  |

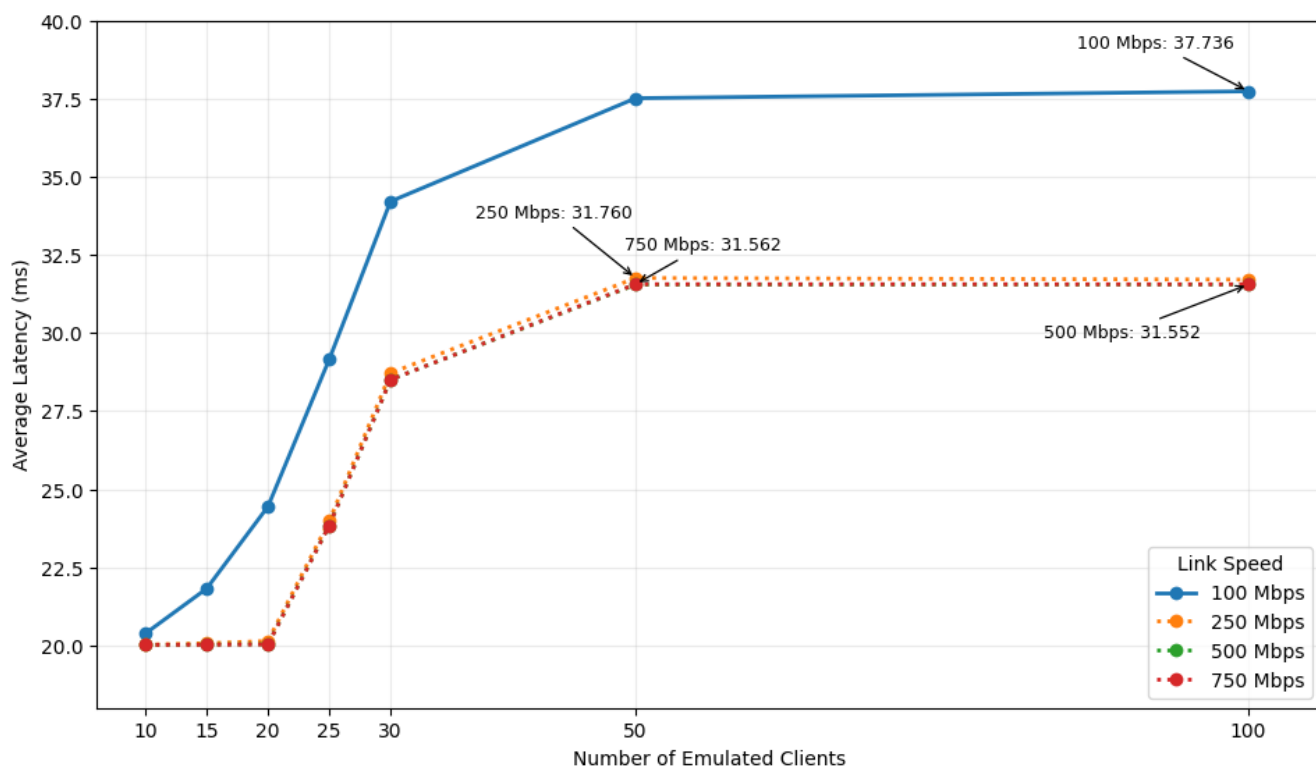
## Prime Throughput vs. Emulated Clients Under Varying Link Speeds



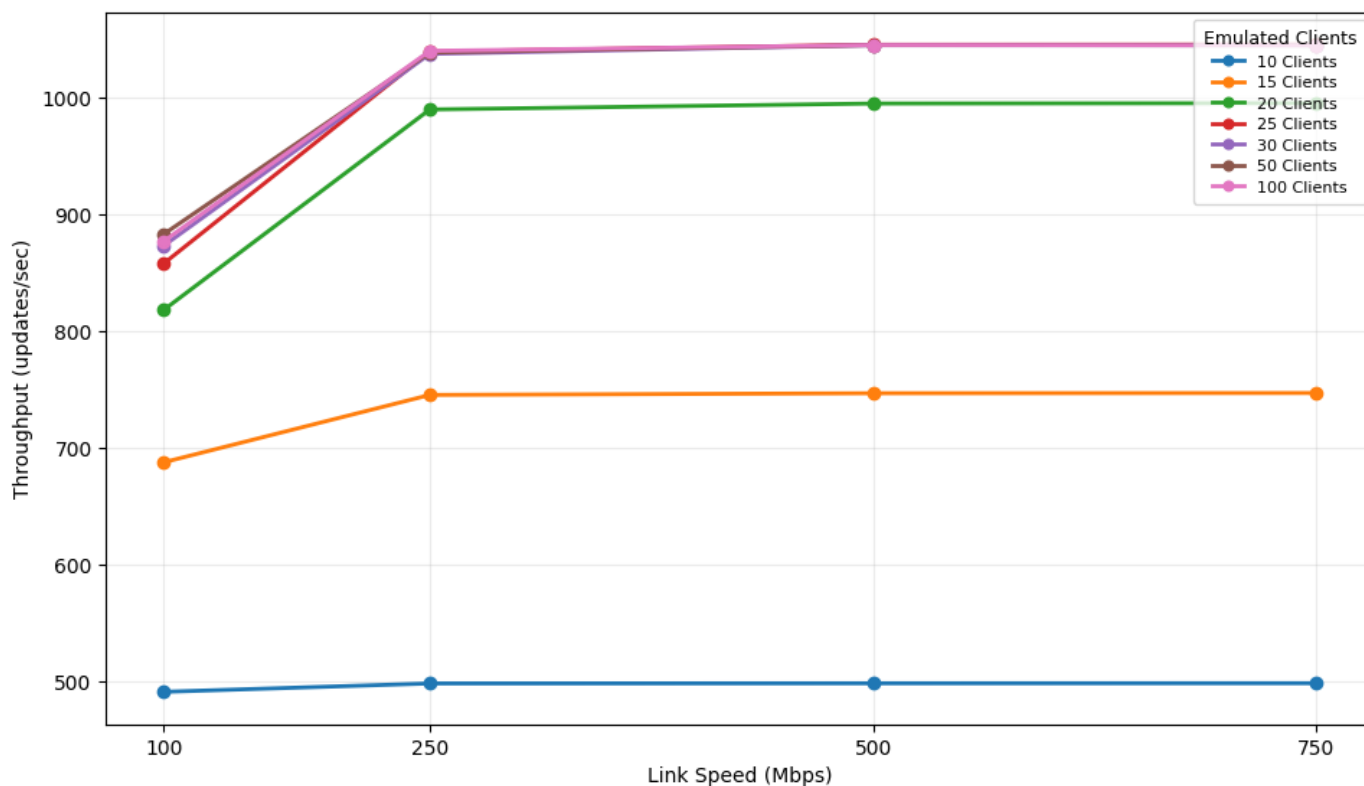
## Prime Average Latency Under Varying Link Speeds



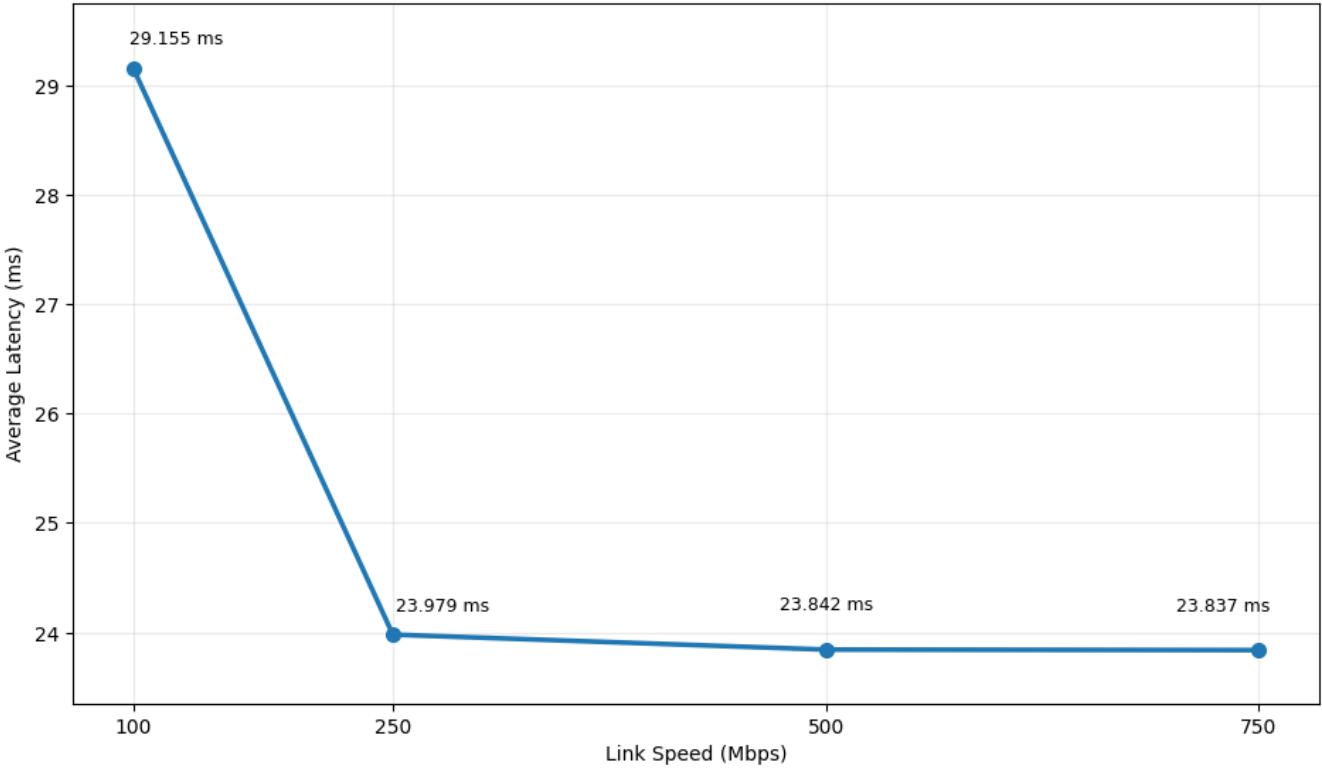
## Prime Average Latency vs. Emulated Clients Under Varying Link Speeds



## Prime Throughput Under Varying Link Speeds



**Prime Average Latency vs. Link Speed at 25 Emulated Clients**



## Multi-Driver Configuration

The multi-driver configuration has a driver program executing on each of the six prime containers. The workload of 25k updates is then split across all containers, or ~4167 updates assigned to each container's driver program.

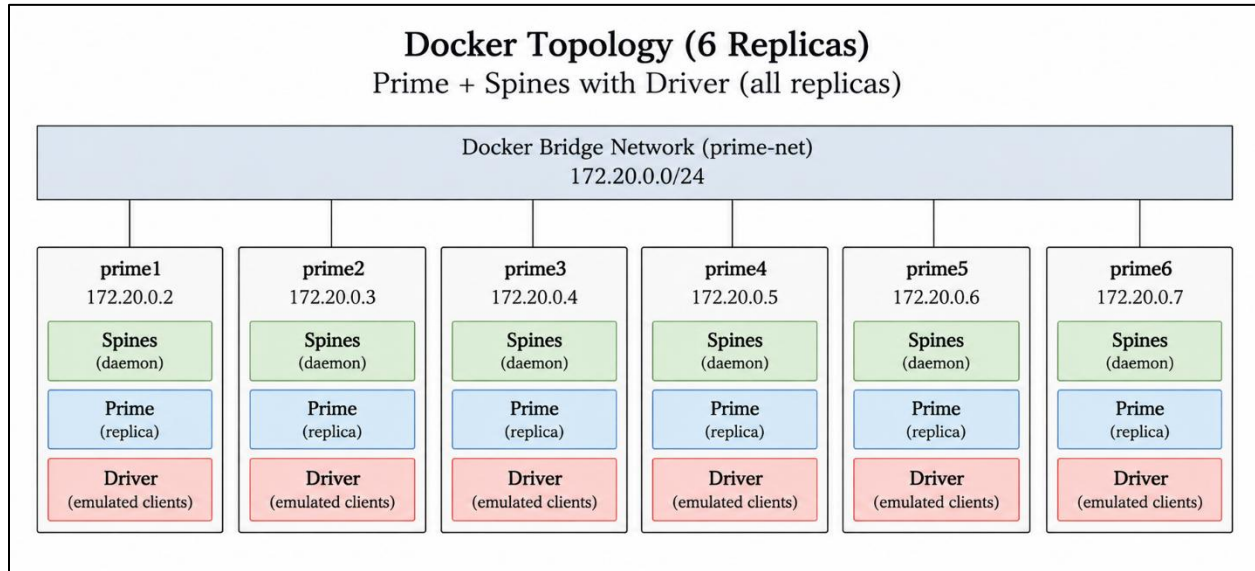


Figure 2. Each container now has its own driver process running and gets assigned its equal share of updates

| 500 Mbps Link Speed (total updates = 25k, number of emulated clients = 25) |                             |      |              |                     |                     |                     |
|----------------------------------------------------------------------------|-----------------------------|------|--------------|---------------------|---------------------|---------------------|
| Container                                                                  | Throughput<br>(updates/sec) | Time | Failed Sends | Min Latency<br>(ms) | Max Latency<br>(ms) | Avg Latency<br>(ms) |
| prime1                                                                     | 1001.61                     | 4.16 | 759          | 18.926              | 43.811              | 23.727              |
| prime2                                                                     | 999.54                      | 4.17 | 726          | 19.116              | 51.909              | 23.775              |
| prime3                                                                     | 998.51                      | 4.17 | 726          | 17.075              | 40.837              | 23.688              |
| prime4                                                                     | 993.47                      | 4.19 | 713          | 18.986              | 40.818              | 23.694              |
| prime5                                                                     | 1001.6                      | 4.16 | 693          | 18.763              | 43.847              | 23.787              |
| prime6                                                                     | 998.79                      | 4.17 | 685          | 16.698              | 40.848              | 23.735              |
| Average                                                                    | 998.92                      | 4.17 | 717          | 18.261              | 43.678              | 23.734              |

When the workload was divided equally amongst all six containers, the updates were processed concurrently in an average of **4.17 seconds**, representing an **82.57% decrease in execution time** compared to running the same workload with a single driver (23.92 seconds).

$$\text{Throughput} = \frac{\text{total updates}}{\text{execution time}} = \frac{25,000 \text{ updates}}{4.17 \text{ seconds}} = 5,996.9 \text{ updates/sec}$$

The resulting throughput represents an increase of roughly **474%** in performance when compared to the same test run using a single driver (1,045.12 updates/sec).

Any link speeds exceeding 500 Mbps did not result in any significant increase in performance. Additionally, anything beyond 25 emulated clients had no significant increase in performance. Further testing is required to confirm the reliability of these results.

## Single-Driver Configuration Raw Data

Below is the raw data from the Single-Driver testing.

| 100 Mbps Link Speed (total updates = 25k) |                          |                     |                    |                        |        |        |
|-------------------------------------------|--------------------------|---------------------|--------------------|------------------------|--------|--------|
| Number of Emulated Clients: 10            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 491.53                   | 50.86               | 0                  | 17.264                 | 42.245 | 20.337 |
| 2                                         | 492.36                   | 50.78               | 0                  | 18.129                 | 49.612 | 20.31  |
| 3                                         | 485.98                   | 51.44               | 0                  | 18.465                 | 42.455 | 20.574 |
| 4                                         | 490.41                   | 50.98               | 0                  | 18.274                 | 40.768 | 20.385 |
| 5                                         | 492.65                   | 50.75               | 0                  | 18.018                 | 41.575 | 20.295 |
| Average                                   | 490.59                   | 50.96               | 0                  | 18.03                  | 43.331 | 20.38  |
| Number of Emulated Clients: 15            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 690.01                   | 36.23               | 2                  | 16.026                 | 41.269 | 21.732 |
| 2                                         | 686.33                   | 36.43               | 0                  | 17.845                 | 41.28  | 21.843 |
| 3                                         | 685.52                   | 36.47               | 0                  | 15.274                 | 40.793 | 21.878 |
| 4                                         | 689.35                   | 36.27               | 0                  | 16.98                  | 43.851 | 21.754 |
| 5                                         | 690.64                   | 36.2                | 2                  | 18.253                 | 41.168 | 21.708 |
| Average                                   | 687.29                   | 36.38               | 0.67               | 16.382                 | 41.114 | 21.818 |
| Number of Emulated Clients: 20            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 818.65                   | 30.54               | 3                  | 17.847                 | 41.084 | 24.419 |
| 2                                         | 824.32                   | 30.33               | 4                  | 17.42                  | 43.469 | 24.258 |
| 3                                         | 813.36                   | 30.74               | 0                  | 16.286                 | 43.687 | 24.578 |
| 4                                         | 815.22                   | 30.67               | 2                  | 18.063                 | 46.664 | 24.518 |
| 5                                         | 817.99                   | 30.56               | 4                  | 16.446                 | 43.760 | 24.447 |
| Average                                   | 817.91                   | 30.57               | 2.6                | 17.212                 | 43.733 | 24.444 |
| Number of Emulated Clients: 25            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 854.14                   | 29.27               | 599                | 18.74                  | 61.742 | 29.232 |
| 2                                         | 856.72                   | 29.18               | 561                | 18.799                 | 59.725 | 29.166 |
| 3                                         | 858.89                   | 29.19               | 575                | 18.618                 | 60.053 | 29.182 |
| 4                                         | 858.34                   | 29.11               | 567                | 17.307                 | 59.99  | 29.086 |
| 5                                         | 860.42                   | 29.13               | 576                | 18.82                  | 59.611 | 29.108 |
| Average                                   | 857.7                    | 29.18               | 575.60             | 18.457                 | 60.224 | 29.155 |
| Number of Emulated Clients: 30            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 864.97                   | 28.9                | 5048               | 18.543                 | 59.94  | 34.516 |
| 2                                         | 868.52                   | 28.78               | 4971               | 18.915                 | 78.522 | 34.394 |
| 3                                         | 868.05                   | 28.8                | 4998               | 19.271                 | 61.109 | 34.394 |
| 4                                         | 892.93                   | 28                  | 5461               | 18.302                 | 77.587 | 33.426 |
| 5                                         | 870.27                   | 28.73               | 5201               | 18.693                 | 62.684 | 34.318 |
| Average                                   | 872.95                   | 28.64               | 5135.8             | 18.745                 | 67.968 | 34.21  |
| Number of Emulated Clients: 50            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 911.32                   | 27.43               | 24971              | 18.275                 | 77.908 | 36.271 |
| 2                                         | 867.57                   | 28.82               | 24971              | 19.313                 | 78.584 | 38.148 |
| 3                                         | 877.56                   | 28.49               | 24966              | 19.144                 | 78.69  | 37.688 |
| 4                                         | 870.98                   | 28.7                | 24959              | 19.168                 | 78.408 | 37.995 |
| 5                                         | 882.82                   | 28.32               | 24976              | 17.722                 | 78.471 | 37.471 |
| Average                                   | 882.85                   | 28.35               | 24968.6            | 18.724                 | 78.412 | 37.515 |
| Number of Emulated Clients: 100           |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 871.77                   | 28.68               | 24969              | 19.279                 | 78.423 | 37.951 |
| 2                                         | 888.2                    | 28.15               | 24969              | 18.902                 | 79.03  | 37.255 |
| 3                                         | 872.73                   | 28.65               | 24962              | 19.547                 | 78.644 | 37.907 |
| 4                                         | 866.12                   | 28.86               | 24960              | 19.699                 | 79.083 | 38.211 |
| 5                                         | 885.37                   | 28.24               | 24965              | 19.579                 | 64.51  | 37.357 |
| Average                                   | 876.84                   | 28.52               | 24965              | 19.401                 | 75.938 | 37.736 |

| 250 Mbps Link Speed (total updates = 25k) |                          |                     |                    |                        |        |        |
|-------------------------------------------|--------------------------|---------------------|--------------------|------------------------|--------|--------|
| Number of Emulated Clients: 10            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 497.85                   | 50.22               | 0                  | 15.638                 | 40.234 | 20.019 |
| 2                                         | 497.47                   | 50.25               | 0                  | 17.566                 | 40.343 | 20.033 |
| 3                                         | 497.71                   | 50.23               | 0                  | 16.933                 | 40.212 | 20.025 |
| 4                                         | 497.91                   | 50.21               | 0                  | 17.266                 | 40.122 | 20.019 |
| 5                                         | 497.97                   | 50.2                | 0                  | 15.825                 | 40.209 | 20.018 |
| Average                                   | 497.78                   | 50.22               | 0                  | 16.646                 | 40.224 | 20.023 |
| Number of Emulated Clients: 15            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 745.86                   | 33.52               | 2                  | 17.844                 | 40.269 | 20.047 |
| 2                                         | 744.61                   | 33.57               | 1                  | 18.316                 | 40.292 | 20.082 |
| 3                                         | 742.84                   | 33.65               | 2                  | 18.558                 | 39.673 | 20.121 |
| 4                                         | 746.26                   | 33.5                | 2                  | 17.837                 | 40.201 | 20.036 |
| 5                                         | 745.47                   | 33.54               | 0                  | 17.961                 | 40.198 | 20.058 |
| Average                                   | 745.01                   | 33.56               | 1.4                | 18.103                 | 40.127 | 20.069 |
| Number of Emulated Clients: 20            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 992.53                   | 25.19               | 8                  | 14.981                 | 40.18  | 20.079 |
| 2                                         | 989.55                   | 25.26               | 3                  | 15.936                 | 44.227 | 20.144 |
| 3                                         | 989.55                   | 25.26               | 2                  | 16.668                 | 40.778 | 20.147 |
| 4                                         | 989.52                   | 25.26               | 6                  | 16.39                  | 40.181 | 20.137 |
| 5                                         | 986.86                   | 25.33               | 1                  | 18.06                  | 46.298 | 20.196 |
| Average                                   | 989.6                    | 25.26               | 4                  | 16.407                 | 42.333 | 20.141 |
| Number of Emulated Clients: 25            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1037.35                  | 24.1                | 1476               | 15.483                 | 41.501 | 24.02  |
| 2                                         | 1044.09                  | 23.94               | 1484               | 18.925                 | 59.03  | 23.867 |
| 3                                         | 1037.53                  | 24.1                | 1470               | 19.208                 | 59.246 | 24.015 |
| 4                                         | 1036.53                  | 24.12               | 1458               | 15.746                 | 59.235 | 24.044 |
| 5                                         | 1040.35                  | 24.03               | 1480               | 17.663                 | 45.95  | 23.949 |
| Average                                   | 1039.17                  | 24.06               | 1473.6             | 17.405                 | 52.992 | 23.979 |
| Number of Emulated Clients: 30            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1036.81                  | 24.11               | 7348               | 18.941                 | 59.516 | 28.729 |
| 2                                         | 1040.25                  | 24.03               | 7405               | 16.602                 | 47.978 | 28.631 |
| 3                                         | 1034.07                  | 24.18               | 7329               | 18.829                 | 59.48  | 28.793 |
| 4                                         | 1036.95                  | 24.11               | 7342               | 18.957                 | 79.789 | 28.73  |
| 5                                         | 1038.12                  | 24.08               | 7375               | 18.849                 | 59.327 | 28.693 |
| Average                                   | 1037.24                  | 24.1                | 7359.8             | 18.436                 | 61.218 | 28.715 |
| Number of Emulated Clients: 50            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1042.52                  | 23.98               | 24968              | 18.059                 | 60.419 | 31.634 |
| 2                                         | 1033.56                  | 24.19               | 24968              | 18.827                 | 61.384 | 31.913 |
| 3                                         | 1037.49                  | 24.1                | 24969              | 18.936                 | 60.365 | 31.791 |
| 4                                         | 1042.29                  | 23.99               | 24969              | 19.104                 | 60.506 | 31.64  |
| 5                                         | 1036.46                  | 24.12               | 24970              | 19.153                 | 90.033 | 31.823 |
| Average                                   | 1038.46                  | 24.08               | 24968.8            | 18.816                 | 66.541 | 31.76  |
| Number of Emulated Clients: 100           |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1037.59                  | 24.09               | 24960              | 17.802                 | 60.21  | 31.773 |
| 2                                         | 1039.91                  | 24.04               | 24974              | 17.204                 | 58.851 | 31.711 |
| 3                                         | 1042.01                  | 23.99               | 24969              | 17.125                 | 60.263 | 31.65  |
| 4                                         | 1038.32                  | 24.08               | 24967              | 14.387                 | 60.432 | 31.746 |
| 5                                         | 1041.24                  | 24.01               | 24960              | 18.981                 | 58.741 | 31.661 |
| Average                                   | 1039.81                  | 24.04               | 24966              | 17.1                   | 59.699 | 31.708 |

| 500 Mbps Link Speed (total updates = 25k) |                          |                     |                    |                        |        |        |
|-------------------------------------------|--------------------------|---------------------|--------------------|------------------------|--------|--------|
| Number of Emulated Clients: 10            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 498.03                   | 50.2                | 0                  | 17.13                  | 33.368 | 20.015 |
| 2                                         | 497.88                   | 50.21               | 0                  | 15.09                  | 39.708 | 20.014 |
| 3                                         | 498.16                   | 50.18               | 0                  | 13.839                 | 40.008 | 20.01  |
| 4                                         | 497.78                   | 50.22               | 0                  | 17.805                 | 39.474 | 20.018 |
| 5                                         | 497.75                   | 50.23               | 0                  | 15.627                 | 40.212 | 20.019 |
| Average                                   | 497.92                   | 50.21               | 0                  | 15.898                 | 38.554 | 20.015 |
| Number of Emulated Clients: 15            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 746.73                   | 33.48               | 0                  | 18.166                 | 40.007 | 20.023 |
| 2                                         | 746.6                    | 33.49               | 0                  | 17.535                 | 39.269 | 20.022 |
| 3                                         | 746.15                   | 33.51               | 0                  | 18.315                 | 39.394 | 20.035 |
| 4                                         | 746.3                    | 33.5                | 0                  | 17.42                  | 39.915 | 20.025 |
| 5                                         | 746.52                   | 33.49               | 1                  | 18.44                  | 40.102 | 20.024 |
| Average                                   | 746.46                   | 33.49               | 0.2                | 17.975                 | 39.737 | 20.026 |
| Number of Emulated Clients: 20            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 994.87                   | 25.13               | 1                  | 17.606                 | 39.332 | 20.036 |
| 2                                         | 994.32                   | 25.14               | 0                  | 18.316                 | 39.454 | 20.051 |
| 3                                         | 994.77                   | 25.13               | 10                 | 18.309                 | 39.127 | 20.035 |
| 4                                         | 995.3                    | 25.12               | 1                  | 17.821                 | 39.163 | 20.027 |
| 5                                         | 994.36                   | 25.14               | 3                  | 16.215                 | 39.318 | 20.046 |
| Average                                   | 994.72                   | 25.13               | 3                  | 17.653                 | 39.279 | 20.039 |
| Number of Emulated Clients: 25            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1044.32                  | 23.94               | 1499               | 18.486                 | 59.374 | 23.863 |
| 2                                         | 1046.04                  | 23.9                | 1491               | 15.815                 | 42.875 | 23.826 |
| 3                                         | 1045.07                  | 23.92               | 1463               | 18.791                 | 40.533 | 23.835 |
| 4                                         | 1044.73                  | 23.93               | 1477               | 18.695                 | 46.526 | 23.846 |
| 5                                         | 1045.46                  | 23.91               | 1474               | 18.468                 | 48.864 | 23.839 |
| Average                                   | 1045.12                  | 23.92               | 1480.8             | 18.051                 | 47.634 | 23.842 |
| Number of Emulated Clients: 30            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1044.55                  | 23.93               | 7431               | 18.052                 | 45.508 | 28.496 |
| 2                                         | 1044.99                  | 23.92               | 7400               | 18.431                 | 59.264 | 28.498 |
| 3                                         | 1045                     | 23.92               | 7419               | 18.607                 | 59.414 | 28.505 |
| 4                                         | 1044.19                  | 23.94               | 7397               | 17.826                 | 58.826 | 28.522 |
| 5                                         | 1044.56                  | 23.93               | 7414               | 19.032                 | 59.026 | 28.508 |
| Average                                   | 1044.66                  | 23.93               | 7412.2             | 18.39                  | 56.408 | 28.506 |
| Number of Emulated Clients: 50            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1045.95                  | 23.9                | 24956              | 18.693                 | 41.777 | 31.521 |
| 2                                         | 1045.33                  | 23.92               | 24965              | 15.734                 | 41.597 | 31.527 |
| 3                                         | 1044.7                   | 23.93               | 24961              | 18.73                  | 49.29  | 31.552 |
| 4                                         | 1043.97                  | 23.95               | 24972              | 19.389                 | 60.299 | 31.566 |
| 5                                         | 1043.88                  | 23.95               | 24959              | 19.211                 | 61.207 | 31.581 |
| Average                                   | 1044.77                  | 23.93               | 24962.6            | 18.351                 | 50.834 | 31.549 |
| Number of Emulated Clients: 100           |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1045.74                  | 23.91               | 24957              | 19.004                 | 43.288 | 31.523 |
| 2                                         | 1045.76                  | 23.91               | 24975              | 17.927                 | 43.723 | 31.528 |
| 3                                         | 1043.78                  | 23.95               | 24976              | 19.116                 | 60.492 | 31.588 |
| 4                                         | 1043.94                  | 23.95               | 24959              | 18.761                 | 60.41  | 31.579 |
| 5                                         | 1045.18                  | 23.92               | 24957              | 17.909                 | 58.602 | 31.544 |
| Average                                   | 1044.88                  | 23.93               | 24964.8            | 18.543                 | 53.303 | 31.552 |

| 750 Mbps Link Speed (total updates = 25k) |                          |                     |                    |                        |        |        |
|-------------------------------------------|--------------------------|---------------------|--------------------|------------------------|--------|--------|
| Number of Emulated Clients: 10            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 498.06                   | 50.19               | 0                  | 15.23                  | 39.648 | 20.013 |
| 2                                         | 497.92                   | 50.21               | 0                  | 14.277                 | 40.455 | 20.017 |
| 3                                         | 497.91                   | 50.21               | 0                  | 17.854                 | 40.122 | 20.019 |
| 4                                         | 498.19                   | 50.18               | 0                  | 16.668                 | 30.952 | 20.008 |
| 5                                         | 497.88                   | 50.21               | 0                  | 18.158                 | 36.338 | 20.021 |
| Average                                   | 497.99                   | 50.2                | 0                  | 16.437                 | 37.503 | 20.016 |
| Number of Emulated Clients: 15            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 746.63                   | 33.48               | 0                  | 18.068                 | 39.97  | 20.019 |
| 2                                         | 747.15                   | 33.46               | 0                  | 18.32                  | 39.249 | 20.013 |
| 3                                         | 746.84                   | 33.47               | 0                  | 17.716                 | 39.188 | 20.015 |
| 4                                         | 746.11                   | 33.51               | 0                  | 18.431                 | 39.816 | 20.039 |
| 5                                         | 746.66                   | 33.48               | 2                  | 16.27                  | 39.453 | 20.017 |
| Average                                   | 746.68                   | 33.48               | 0.4                | 17.761                 | 39.535 | 20.021 |
| Number of Emulated Clients: 20            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 995.51                   | 25.11               | 2                  | 17.417                 | 39.116 | 20.018 |
| 2                                         | 995.01                   | 25.13               | 3                  | 16.388                 | 39.191 | 20.031 |
| 3                                         | 995.77                   | 25.11               | 10                 | 18.11                  | 39.273 | 20.014 |
| 4                                         | 995.21                   | 25.12               | 9                  | 14.668                 | 40.062 | 20.027 |
| 5                                         | 993.7                    | 25.16               | 7                  | 17.773                 | 39.479 | 20.064 |
| Average                                   | 995.04                   | 25.13               | 6.2                | 16.871                 | 39.424 | 20.031 |
| Number of Emulated Clients: 25            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1045.49                  | 23.91               | 1506               | 18.881                 | 48.084 | 23.836 |
| 2                                         | 1045.82                  | 23.9                | 1485               | 17.123                 | 41.827 | 23.827 |
| 3                                         | 1045.06                  | 23.92               | 1480               | 18.749                 | 40.737 | 23.838 |
| 4                                         | 1044.96                  | 23.92               | 1456               | 18.268                 | 59.158 | 23.842 |
| 5                                         | 1045.15                  | 23.92               | 1458               | 18.327                 | 40.892 | 23.843 |
| Average                                   | 1045.3                   | 23.91               | 1477               | 18.27                  | 46.14  | 23.837 |
| Number of Emulated Clients: 30            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1045.05                  | 23.92               | 7420               | 19.024                 | 58.877 | 28.502 |
| 2                                         | 1044.9                   | 23.93               | 7408               | 18.302                 | 41.464 | 28.497 |
| 3                                         | 1043.67                  | 23.95               | 7409               | 18.868                 | 58.873 | 28.525 |
| 4                                         | 1044.19                  | 23.94               | 7437               | 18.229                 | 60.305 | 28.517 |
| 5                                         | 1045.48                  | 23.91               | 7424               | 18.639                 | 49.412 | 28.489 |
| Average                                   | 1044.66                  | 23.93               | 7419.6             | 18.612                 | 53.786 | 28.506 |
| Number of Emulated Clients: 50            |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1043.44                  | 23.96               | 24974              | 16.753                 | 59.548 | 31.6   |
| 2                                         | 1045.39                  | 23.91               | 24976              | 19.167                 | 49.897 | 31.538 |
| 3                                         | 1044.68                  | 23.93               | 24979              | 19.184                 | 58.863 |        |
| 4                                         | 1045.54                  | 23.91               | 24968              | 18.08                  | 47.653 | 31.539 |
| 5                                         | 1044.4                   | 23.94               | 24973              | 19.078                 | 60.291 | 31.57  |
| Average                                   | 1044.69                  | 23.93               | 24974              | 18.452                 | 55.25  | 31.562 |
| Number of Emulated Clients: 100           |                          |                     |                    | Latency (milliseconds) |        |        |
| Trial Number                              | Throughput (updates/sec) | Time Elapsed (secs) | Total Failed Sends | min                    | max    | avg    |
| 1                                         | 1044.54                  | 23.93               | 24974              | 19.295                 | 58.807 | 31.566 |
| 2                                         | 1045.38                  | 23.91               | 24973              | 7.592                  | 60.808 | 31.541 |
| 3                                         | 1044.76                  | 23.93               | 24962              | 17.878                 | 58.933 | 31.554 |
| 4                                         | 1044.35                  | 23.94               | 24964              | 17.838                 | 58.97  | 31.56  |
| 5                                         | 1043.94                  | 23.95               | 24961              | 18.998                 | 60.334 | 31.576 |
| Average                                   | 1044.59                  | 23.93               | 24966.8            | 16.32                  | 59.57  | 31.559 |